

Atul Kumar

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Profile

Software Developer with hands-on experience building full-stack web applications and AI-powered systems using Python, JavaScript, and modern frameworks like Next.js and Flask. Comfortable across the stack, from REST API design and database modeling to responsive frontend development. Quick learner with strong problem-solving, communication, and teamwork abilities.

Education

Godavari Institute of Engineering and Technology 2023 – 2027
B.Tech in Electronics and Communication Engineering (CGPA: 7.5)

Govt Higher Secondary School, Garh Dhamma 2021 – 2023
12th (80%)

Experience

Intern – Embedded Systems / IoT, Genz Educate Wing Feb 2026 – May 2026

- Built the software logic for an Adaptive Smart Traffic Signal Controller, implementing FSM-based control flow and real-time simulation using Wokwi Simulator.
- Collaborated on sensor integration and system testing in an agile, fast-paced team environment.

Projects

SensorNest – Smart Device Monitoring Platform [View Live →](#) *Next.js, React, Node.js, MongoDB, Python*

- Built a full-stack web platform with a Next.js frontend, REST APIs, and a MongoDB backend to ingest and visualize real-time device telemetry.
- Implemented secure user authentication, role-based access, and TTL-indexed MongoDB storage for automatic 7-day data retention.
- Programmed a Python bridge service to relay live sensor data from an IoT device's local IP to the cloud-hosted backend.

PlantCare AI – Plant Disease Monitoring System [View Live →](#) *Python, TensorFlow/Keras, Flask, Next.js, OpenCV*

- Trained a custom CNN (TensorFlow/Keras) to classify plant leaf diseases across 71 crop-disease categories, and designed a 2-stage hierarchical inference pipeline to eliminate cross-crop false positives.
- Built a Flask REST API and a Next.js frontend enabling users to upload images and instantly view diagnosis results.

Smart AI-Powered Traffic Signal Controller [View Live →](#) *Python, YOLOv8, OpenCV, Flask, React*

- Built a real-time vehicle detection pipeline (YOLOv8, OpenCV) and an adaptive signal-timing algorithm (10s–60s) driven by live traffic density.
- Designed a Flask REST API and a React dashboard for live video streaming and traffic statistics.

Skills

- **Programming:** Python, JavaScript, C, C++ (Basics)
- **Web / Full-Stack:** Next.js, React, Node.js, MongoDB, Flask, REST APIs
- **Tools:** Git (Basics), VS Code
- **Other:** ESP32 / IoT Basics, Arduino IDE
- **Soft Skills:** Problem Solving, Communication, Teamwork, Adaptability
- **Languages:** English, Hindi, Telugu

Certifications & Achievements

- Blockchain and Applications, Prompt Engineering, Python Programming – Great Learning
- Cloud Fundamentals – Skill Camp
- Head Boy, Govt Higher Secondary School Garh Dhamma (2021–2023) — PMSSS Scholarship Recipient